

LOUCA MALERBA

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EDUCATION

CentraleSupélec, Université Paris-Saclay 2023 – 2026
Top 3 Engineering School in France, part of 13th-ranked university worldwide (Shanghai Ranking)
Master's Degree (M2) in General Engineering - Data Science Specialization — GPA: 4.04/4.33

Coursework: Machine Learning, Deep Learning, Optimization, Statistical Modeling, Statistical Learning, Representation Learning, Big Data Computing, Reinforcement Learning, NLP, Sparse Models, HPC

CPGE Jules Ferry 2021 – 2023
Preparatory Classes - PTSI/PT - Ranked 5th nationwide - Top 5% of students*

Intensive foundation program in mathematics, physics, engineering — National competitive exam preparation

EXPERIENCE

GE HealthCare, Buc, France Oct 2023 – Sept 2026 (Expected)
Apprentice Research Engineer

- End-to-end deep learning pipeline for chronic pulmonary embolism (CTEPH) segmentation and classification in contrast-enhanced CT angiography as part of *Destination 2024* partnership between GE HealthCare and AP-HP, from literature review to clinical validation on hospital data warehouses. (*link to poster*)
- Identified data leakage in the RSPECT public reference dataset and conducted the first lesion-level classification study on an expert-annotated clinical dataset for this pathology, achieving performance significantly above what is achievable on the public benchmark.
- **Targeting a publication** in a peer-reviewed venue in medical imaging

European Leader in LLMs May 2026 – Present
Data Annotator

- Part-time (10 hours/week) data annotation work for a European Leader in LLM (details confidential under NDA).

PROJECTS

Dual-sPLS: a family of Dual Sparse Partial Least Squares regressions for feature selection and prediction with tunable sparsity (*link*)

Independent Research Collaboration with Laurent Duval, Adjunct Professor at CentraleSupélec

- Acknowledged contribution in forthcoming paper (CalValXy); Contributor of work in progress (Dual-sPLS)

Active Learning for Plankton Segmentation — Side Project (*link*)

- Human-in-the-loop pipeline to segment images under significant domain shift and uncertainty sampling strategy with stochastic batches

*More projects are in progress or available on my portfolio (*link*)*

TECHNICAL SKILLS

Languages, Software Engineering & Infrastructure: Python, C++, Git, Linux, SLURM

Deep Learning & ML: PyTorch, Scikit-learn

Computer Vision for Medical AI: nnUNet, nnDetection, SimpleITK, 3D Slicer

LANGUAGES

French: Native — **English:** Fluent (C1 – TOEFL 111/120) — **Italian:** Intermediate